James Pritts

https://prittjam.github.io/

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Academic Appointments

Christian-Albrechts-Universität zu Kiel — Kiel, Germany — 2024 – Now
Marie Skłodowska-Curie (MSCA) Fellow

Ukrainian Catholic University — Lviv, Ukraine — 2023 – 2024
Assistant Professor

Czech Inst. of Informatics, Robotics & Cybernetics — Prague, Czechia — 2022 – 2023
Research Fellow

Chalmers University of Technology — Gothenburg, Sweden — 2021
Post-Doctoral Research Scientist

Facebook Reality Labs, AR/VR — Pittsburgh, PA — 2019 – 2021
Post-Doctoral Research Scientist

Education

Czech Technical University — Prague, Czechia — 2020
Ph.D., Computer Science
Thesis: "Methods for the Rectification of Imaged Coplanar Repeated Patterns"

Czech Technical University — Prague, Czechia — 2013
M.Sc., Computer Science

The University of North Texas — Denton, TX — 2002
B.Sc., Mathematics

Publications

J. Pritts and K. Köser. Robust Local Optimization Done Right. In *ACCV*, 2026

J. Pritts, F. Seegräber, and K. Köser. RANSAC Scoring Done Right. In *NeurIPS*, 2026

J. Pritts, T. Sittart, H. Sauer, S. Janßen, F. Seegräber, D. Nakath, and K. Köser. BlobBoards: Robust Markers for Accurate Pose. In *BMVC*, 2026

E. Dexheimer, P. Peluse, J. Chen, **J. Pritts**, and M. Kaess. Information-theoretic online multi-camera extrinsic calibration. *IEEE Robotics and Automation Letters*, 7(2):4757–4764, 2022

Y. Lochman, K. Liepieshov, J. Chen, M. Perdoch, C. Zach, and **J. Pritts**. Babelcalib: a universal approach to calibrating central cameras. In *ICCV*, 2021

Y. Lochman, O. Dobosevych, R. Hryniv, and **J. Pritts**. Minimal Solvers for Single-View Lens-Distorted Camera Auto-Calibration. In *WACV*, 2021

J. Pritts, Z. Kukelova, V. Larsson, Y. Lochman, and O. Chum. Minimal solvers for rectifying from radially-distorted conjugate translations. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(11):3931–3948, 2021

J. Pritts, Z. Kukelova, V. Larsson, Y. Lochman, and O. Chum. Minimal Solvers for Rectifying from Radially-Distorted Scales and Change of Scales. *International Journal of Computer Vision*, 128(4):950–968, 2020

J. Pritts, Z. Kukelova, V. Larsson, and O. Chum. Rectification from Radially-Distorted Scales. In *ACCV*, 2018

J. Pritts, Z. Kukelova, V. Larsson, and O. Chum. Radially-Distorted Conjugate Translations. In *CVPR*, 2018

J. Pritts, D. Rozumnyi, M. P. Kumar, and O. Chum. Coplanar Repeats by Energy Minimization. In *BMVC*, 2016

J. Pritts, O. Chum, and J. Matas. Detection, Rectification and Segmentation of Coplanar Repeated Patterns. In *CVPR*, 2014

J. Pritts, O. Chum, and J. Matas. Approximate Models for Fast and Accurate Epipolar Geometry Estimation. In *IVCNZ*, 2013

Software

VisualGeometryToolkit.jl 2026
Julia packages for multiple-view geometry: primitives, features, robust estimation, RANSAC scoring and evaluation. DOI: 10.5281/zenodo.23548372

BlobBoards 2026
Application for blob marker boards: authoring printable boards, detecting and identifying them in images, and estimating their pose.

Recognition and Awards

Asian Conference on Computer Vision (ACCV) Saburo Tsuji Best Paper Award 2018
for "Rectification from Radially-Distorted Scales"

Image and Vision Computing New Zealand (IVCVNZ) Best Paper Award 2013
for "Approximate Models for Fast and Accurate Epipolar Geometry Estimation"

Professional Association Memberships

ELLIS	Scholar	2024 – Now
Kiel Marine Science	Associate	2025 – Now

Funding

Principal	Kiel Marine Science (KMS) "Enhancing Underwater Robotics for Visual Mapping in Turbid Environments" Awarded: €10,000	2025
Principal	Marie Skłodowska-Curie Actions (MSCA) Fellowship, via Kiel Training for Excellence (KiTE) "Live and Robust Visual Localization and Mapping in Coastal Waters" Awarded: $225,000	2024 – 2027
Contributing	Facebook Sponsored Research Agreement "In-the-field Extrinsic Calibration of Multi-camera Systems" Awarded: $150,000	2020 – 2021
Principal	Facebook Sponsored Research Agreement "Calibration of Head-Mounted Multi-Camera Capture Systems" Awarded: $25,000	2020

Professional Service Activities

Reviewer	CVPR, ICCV, ECCV, 3DV, WACV, ACCV, ICRA, IJCV, PAMI	
Conference Organizer	Poster Chair	3DV 2022
Tutorial Organizer	"Affine Correspondences and Their Applications"	3DV 2022
Tutorial Organizer	"Affine Correspondences and Their Applications"	CVPR 2022

Teaching

Kiel University

3D Computer Vision — 2025 – 2026
Substitute Instructor, M.Sc. level

Ukrainian Catholic University

3D Computer Vision — 2023
Instructor, M.Sc. level

Computer Vision — 2022
Guest Lecturer, M.Sc. level

Image Retrieval — 2017
Instructor, M.Sc. level

Czech Technical University in Prague

Pattern Recognition and Machine Learning — 2013 – 2016
Teaching Assistant, B.Sc. level

Advising

M.Sc. Students

Philipp Böll	Kiel University "Underwater Light Rig Auto-Calibration from Asynchronous Cameras"	in progress
Ties Harder	Kiel University "Boundary Movement Aware Differentiable Rendering for Visual Calibration Tasks"	2026
Hendrik Sauer	Kiel University "Affine-Equivariant Neural Networks for the Feature Matching Problem"	2026
Till Sittart	Kiel University "Affine-Equivariant Blob Targets for Robust Absolute Pose in Robotics"	2026
Liam Boddin	Kiel University "Underwater Calibration of Near-point Anisotropic Lights"	2025
Igor Babin	Ukrainian Catholic University "Image Inpainting in Latent Space"	2024
Dmytro Nadobko	Ukrainian Catholic University "Supervised Learning of Correspondence Volumes for Coplanar Repetitive Patterns"	2023
Andrii Stadnik	Ukrainian Catholic University "Corner Localization and Camera Calibration from Imaged Lattices"	2023
Yaroslava Lochman	Ukrainian Catholic University "Minimal Solvers for Single-View Auto-Calibration"	2020

B.Sc. Students

Yaroslav Romanus	Ukrainian Catholic University "Applying Motion in Latent Space"	2024
Ostap Viniavskyi	Ukrainian Catholic University "Learning Discriminative Context-Aware Keypoints Representations for Resolving Ambiguous Matches"	2021
Kostiantyn Liepieshov	Ukrainian Catholic University "Manhattan-Frame Detection in Lens-Distorted Images"	2021

Invited Talks and Lectures

Fishbowl Discussion, Annual Kiel Marine Science Retreat "NextGen Perspectives"	September 2026 Kiel, Germany
3DV Tutorial Talk "Just One Image is All It Takes..."	September 12, 2022 Prague, Czechia
CVPR Tutorial Talk "Just One Image is All It Takes..."	June 19, 2022 New Orleans, LA
Facebook Reality Labs "Minimal Solvers and Multi-Model Estimators for Image Undistortion and Scene-Plane Rectification"	June 14, 2018 Pittsburgh, PA
The Aspen Institute's 2018 Young Leader's Program "Opportunities and Risks of Artificial Intelligence"	March 17, 2018 Tále, Slovakia
The Eastern European Computer Vision Conference "Detection, Rectification, and Segmentation of Coplanar Repeated Patterns"	July 14, 2017 Odesa, Ukraine
The 34th Pattern Recognition and Computer Vision Colloquium "Detection, Rectification, and Segmentation of Coplanar Repeated Patterns"	April 3, 2014 Prague, Czechia